# Topic -2 (Part 2): Multi-armed Bandits

**AIMLCZG512-Deep Reinforcement Learning**

Lecture Notes

## 1. Topics Covered

1. Stationary and nonstationary reward distributions in bandit problems.
2. Incremental computation of action-value estimates from observed rewards.
3. Step-size choices for sample-average estimation and tracking changing rewards.
4. Optimistic initial values as a simple exploration mechanism.
5. Upper-confidence-bound action selection and its exploration bonus.
6. Gradient bandit algorithms as an alternative to value-based action selection.
7. Contextual bandits as associative search between bandits and full reinforcement learning.
8. Summary of action-selection methods, review questions, and required reading.

## 2. Stationary and Nonstationary Rewards

In a bandit problem, each action has its own reward distribution. The distribution specifies what rewards are likely when that action is selected. The important question is whether this reward distribution remains the same over time, or changes as interaction continues.

**Stationary reward distribution.** A reward distribution is stationary if, for each action $a$, the probability distribution generating rewards for that action does not change over time. In this case the true action value $q_*(a)$ is fixed, and repeated sampling can make the estimate $Q_t(a)$ more reliable.

**Nonstationary reward distribution.** A reward distribution is nonstationary if the probability distribution for one or more actions changes over time. Then the best action may also change as the reward distributions change. It is useful to think of the true value as time-dependent, say $q_t^*(a)$, rather than as one fixed number.

### 2.1. Stationary vs. Nonstationary - With Examples

| Setting | Stationary version | Nonstationary version |
|---|---|---|
| Slot-machine style bandit | Each arm has a fixed payout distribution throughout the experiment. A long sequence of plays gives better estimates of fixed action values. | The payout pattern changes after maintenance, a rule change, or a change in operating condition. Older observations may no longer represent the present arm values. |
| Online advertisement | During a short A/B experiment, the probability of click for each advertisement is assumed nearly fixed. Each advertisement can be treated as having a stable average reward. | User attention changes with season, festival period, repeated exposure, competitor campaigns, or current news. An advertisement that worked well earlier may later become less effective. |
| Treatment choice | Suppose a controlled study compares a few treatment protocols for a clearly defined patient group over a short period. If patient characteristics and clinical conditions are stable, each treatment may be viewed as having an approximately fixed outcome distribution. | In practice, the mix of patients, disease variants, co-medications, adherence, and hospital practice can change. Then the observed outcome distribution of a treatment may drift, so older evidence should not dominate forever. |
| Cloud configuration | A service is tested under a stable workload and fixed infrastructure. Each configuration has a roughly stable distribution of throughput or latency. | Workload, traffic mix, background jobs, network congestion, and hardware contention vary across time. A configuration that was best in the morning may not be best during peak load. |

**In real deployments, it is often safer to expect some degree of nonstationarity.** User populations change, systems are upgraded, demand fluctuates, and external conditions shift. Therefore, practical bandit methods often need to **track** action values rather than merely converge once and stop adapting.

## 3. Incremental Computation of Action Values

Suppose we focus on one action and consider only the rewards obtained after selecting that action. Let $R_1, R_2, \ldots, R_n$ be the first $n$ rewards observed from this action. Let $Q_n$ denote the estimate before observing $R_n$, that is, the average of the first $n-1$ rewards. The new estimate after observing $R_n$ is $Q_{n+1}$.

The direct sample average is

$$Q_{n+1} \doteq \frac{1}{n}\sum_{i=1}^{n} R_i.$$

The same quantity can be computed incrementally, as in the standard derivation of the sample-average update:

$$
\begin{aligned}
Q_{n+1} &= \frac{1}{n}\sum_{i=1}^{n} R_i && \text{average of all } n \text{ observed rewards} \\
&= \frac{1}{n}\left(R_n + \sum_{i=1}^{n-1} R_i\right) && \text{separate the newest reward } R_n \\
&= \frac{1}{n}\left(R_n + (n-1)\frac{1}{n-1}\sum_{i=1}^{n-1} R_i\right) && \text{write the old sum through its average} \\
&= \frac{1}{n}\left(R_n + (n-1)Q_n\right) && \text{replace the previous average by } Q_n \\
&= \frac{1}{n}\left(R_n + nQ_n - Q_n\right) && \text{expand } (n-1)Q_n \text{ as } nQ_n - Q_n \\
&= Q_n + \frac{1}{n}\left[R_n - Q_n\right] && \text{old estimate plus a correction.}
\end{aligned}
$$

This final form is important because the learner does not need to store all earlier rewards. It only needs the current estimate $Q_n$ and the count $n$. Each new reward produces a small correction to the old estimate.

## 4. The General Update Form

The incremental update has a form that appears repeatedly in reinforcement learning:

$$\textbf{NewEstimate} \leftarrow \textbf{OldEstimate} + \textbf{StepSize}\,[\textbf{Target} - \textbf{OldEstimate}].$$

For the sample-average bandit estimate,

$$\textbf{OldEstimate} = Q_n, \qquad \textbf{Target} = R_n, \qquad \textbf{StepSize} = \frac{1}{n}.$$

The term [**Target** − **OldEstimate**] is an estimation error. The **StepSize** determines how far the estimate moves toward the target. A small step-size gives stable but slow change. A large step-size gives faster movement but greater sensitivity to noisy rewards.

### 4.1. Numerical Illustration

Suppose the current estimate for an action is $Q_4 = 3.0$, and the next observed reward is $R_4 = 7.0$. Since this is the fourth reward for this action, the sample-average step-size is $1/4$:

$$Q_5 = 3.0 + \frac{1}{4}(7.0 - 3.0) = 4.0.$$

Now suppose the same reward target 7.0 is observed again. The step-size shrinks as the count grows:

$$Q_6 = 4.0 + \frac{1}{5}(7.0 - 4.0) = 4.6,$$

$$Q_7 = 4.6 + \frac{1}{6}(7.0 - 4.6) = 5.0.$$

The estimate moves toward the target, but each later correction is smaller because the sample-average step-size $1/n$ becomes smaller.

## 5. Step-size Choices and Tracking

The step-size determines how strongly the newest reward affects the estimate. The two basic choices in this part of the topic are sample-average step-size and constant step-size.

| Choice | Update rule | Interpretation |
|---|---|---|
| Sample-average step-size | $Q_{n+1} = Q_n + \frac{1}{n}[R_n - Q_n]$ | Each reward for the action contributes equally to the average. This is natural when the reward distribution is stationary. |
| Constant step-size | $Q_{n+1} = Q_n + \alpha[R_n - Q_n]$ | Recent rewards keep receiving noticeable influence. This helps the estimate track a changing reward distribution. |

### 5.1. Stationary Case

In the stationary case, the true value of an action does not change. Then the sample-average method is natural: as the action is selected repeatedly, $Q_n$ averages more samples and becomes more stable. The step-size $1/n$ becomes smaller over time, so later rewards produce smaller corrections.

### 5.2. Nonstationary Case

In a nonstationary case, old rewards may describe an earlier environment rather than the present one. If the step-size becomes too small, the estimate may adapt too slowly. A constant step-size $\alpha$ avoids this by keeping the estimate responsive to recent rewards. More elaborate time-varying step-sizes may be designed, but they should not decay so quickly that the learner stops adapting to current rewards.

### 5.3. A Simple Tracking Illustration

**Case 1: reward level increases.** Suppose $Q = 10$ and $\alpha = 0.2$. If the next rewards are 14 and 16, then

$$Q \leftarrow 10 + 0.2(14 - 10) = 10.8,$$

$$Q \leftarrow 10.8 + 0.2(16 - 10.8) = 11.84.$$

The estimate begins moving upward, which is appropriate if the action's reward distribution has improved.

**Case 2: reward level decreases.** Suppose $Q = 12$ and $\alpha = 0.2$. If the next rewards are 8 and 7, then

$$Q \leftarrow 12 + 0.2(8 - 12) = 11.2,$$

$$Q \leftarrow 11.2 + 0.2(7 - 11.2) = 10.36.$$

The estimate moves downward, which is useful when older high rewards no longer represent current performance.

## 6. Optimistic Initial Values

Optimistic initial values are a simple way to encourage exploration. Instead of starting all action-value estimates at a neutral or realistic value, the learner starts them higher than the rewards it is likely to observe.

### 6.1. How Optimism Encourages Exploration

1. Initialize all actions optimistically, for example $Q_1(a) = 5$ for every action. *(Every action initially appears attractive.)*
2. Select greedily according to the current estimates. *(No explicit random exploration is required.)*
3. The selected action usually receives a reward below its optimistic estimate. *(The learner is disappointed relative to its initial belief.)*
4. Update that selected action downward. *(Its estimate becomes less attractive than estimates of untried actions.)*
5. Greedy selection is then pulled toward other still-optimistic actions. *(Several actions are tried because untried actions remain overestimated.)*

### 6.2. Why It Works and Why It Is Limited

Optimism creates temporary exploration pressure. Actions that have not yet been tried still look promising, while actions tried once or twice are corrected toward ordinary reward levels. Thus, even greedy action selection can explore early.

The limitation is that this pressure is mostly an initial-condition effect. Once estimates have been corrected, the method does not provide a continuing reason to explore. It can be useful in simple stationary tasks, but it is not a general exploration method for changing environments.

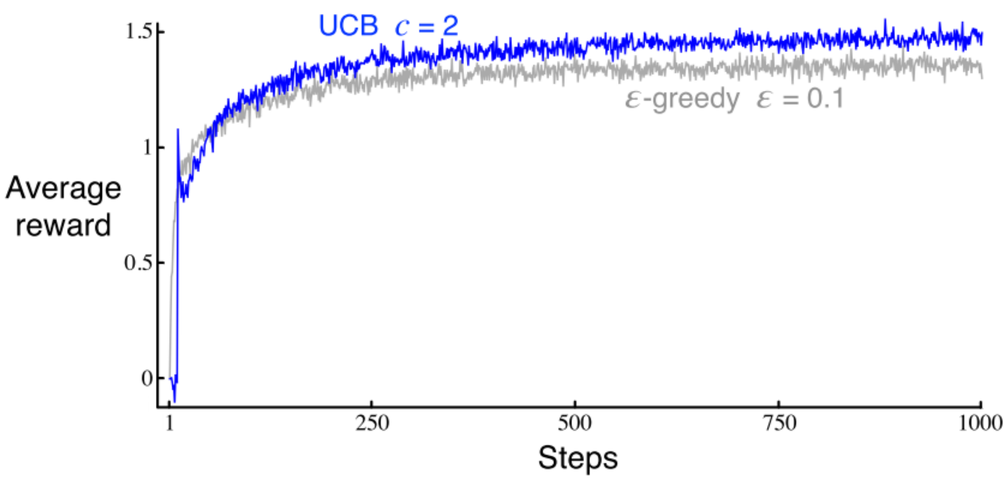
## 7. Upper-Confidence-Bound Action Selection

The upper-confidence-bound, or UCB, method selects actions by combining the current estimated value with an exploration bonus. The method prefers actions that either look good or have not yet been tried enough.

**UCB action selection.**

$$A_t \doteq \arg\max_a \left[ Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}} \right].$$

Here $Q_t(a)$ is the current estimate of action $a$, $N_t(a)$ is the number of times action $a$ has been selected before time $t$, and $c > 0$ controls the degree of exploration. If $N_t(a) = 0$, the action is treated as a maximizing candidate so that untried actions are selected.




Average reward comparison of UCB action selection and $\epsilon$-greedy action selection on the 10-armed testbed.

### 7.1. Intuition Behind the Expression

The first term, $Q_t(a)$, supports exploitation: actions with high current estimated reward look attractive. The second term,

$$c\sqrt{\frac{\ln t}{N_t(a)}},$$

is the exploration bonus. It acts as a measure of uncertainty in the estimate of action $a$'s value. The bonus is large when an action has been selected only a few times. Each time that action is selected, $N_t(a)$ increases and the bonus decreases. Each time some other action is selected, $t$ increases while $N_t(a)$ does not, so the less-sampled action again becomes more competitive.

The parameter $c$ controls the degree of exploration. A larger $c$ gives more weight to the uncertainty term. A smaller $c$ makes the method closer to greedy action selection.

### 7.2. Numerical Examples on UCB

#### Example 1: Two actions

At time $t = 20$, suppose

$$Q_t(a_1) = 1.0,\ N_t(a_1) = 10, \qquad Q_t(a_2) = 0.8,\ N_t(a_2) = 2,$$

and let $c = 2$. The UCB scores are

$$\text{score}(a_1) = 1.0 + 2\sqrt{\frac{\ln 20}{10}} \approx 1.0 + 1.095 = 2.095,$$

$$\text{score}(a_2) = 0.8 + 2\sqrt{\frac{\ln 20}{2}} \approx 0.8 + 2.448 = 3.248.$$

The method selects $a_2$ even though its current estimate is lower, because it has been tried much less often. If the reward obtained from $a_2$ is $R = 1.4$, then the sample-average update after its third selection is

$$Q(a_2) \leftarrow 0.8 + \frac{1}{3}(1.4 - 0.8) = 1.0.$$

Thus, UCB selects the action using an uncertainty-aware score, while the value estimate is still updated using the reward observed from the selected action.

#### Example 2: Four actions

At time $t = 50$, let $c = 1$ and suppose

| Action | $a_1$ | $a_2$ | $a_3$ | $a_4$ |
|---|---|---|---|---|
| $Q_t(a)$ | 1.10 | 1.20 | 0.70 | 0.90 |
| $N_t(a)$ | 25 | 20 | 4 | 1 |

The scores are approximately

| Action | $a_1$ | $a_2$ | $a_3$ | $a_4$ |
|---|---|---|---|---|
| $Q_t(a) + \sqrt{\ln(50)/N_t(a)}$ | 1.50 | 1.64 | 1.69 | 2.88 |

UCB selects $a_4$. Although $a_4$ is not greedy according to $Q_t(a)$ alone, it has been selected only once, so its uncertainty term is large.

## 8. Summary of Action-Selection Methods

| Method | Rule and parameters involved | Exploration behaviour and suitable use |
|---|---|---|
| Greedy | $A_t = \arg\max_a Q_t(a)$; no exploration parameter. | Pure exploitation of current estimates. Useful as a baseline, but can settle on a suboptimal action after misleading early rewards. |
| $\epsilon$-greedy | Select $\arg\max_a Q_t(a)$ with probability $1-\epsilon$; select randomly from all $k$ actions with probability $\epsilon$. Parameter: $\epsilon$. | Simple continuing exploration. Larger $\epsilon$ explores more often and sacrifices more immediate reward; smaller $\epsilon$ exploits more often. |
| Optimistic initial values | Set $Q_1(a)$ high for all actions and then often act greedily. Parameter: initial value $Q_1(a)$. | Encourages early exploration because untried actions remain attractive. Most suitable when an initial burst of exploration is enough. |
| UCB | $A_t = \arg\max_a \left[Q_t(a) + c\sqrt{\ln t/N_t(a)}\right]$. Parameter: $c$. | Selects actions using both estimated value and uncertainty. It favours actions that are close to greedy or have been selected fewer times; the extension beyond bandits is less direct. |

**Choosing between methods.** There is no single best rule independent of the task. Greedy is mainly a baseline. $\epsilon$-greedy is simple and robust when continuing exploration is desired. Optimistic initialization is useful when early exploration is valuable and reasonable optimistic estimates can be chosen. UCB is more directed because it prefers actions with high estimates or high uncertainty, but it relies on meaningful action counts and is most natural in bandit settings.

## 9. (Optional Topic) Gradient Bandit Algorithms

**Instructor note.** This section is optional for this lecture note. It can be read lightly now and revisited later when policy-gradient and actor-critic methods are introduced.

The methods so far estimate action values and then choose actions from those estimates. A gradient bandit method takes a different view. It learns a numerical **preference** $H_t(a)$ for each action. A larger preference means the action is more likely to be selected, but the preference itself is not an estimate of expected reward.

Action probabilities are computed using a softmax distribution:

$$\Pr\{A_t = a\} \doteq \frac{e^{H_t(a)}}{\sum_{b=1}^{k} e^{H_t(b)}} \doteq \pi_t(a).$$

After action $A_t$ is selected and reward $R_t$ is observed, the preferences are updated as

$$H_{t+1}(A_t) = H_t(A_t) + \alpha(R_t - \bar{R}_t)(1 - \pi_t(A_t)),$$

$$H_{t+1}(a) = H_t(a) - \alpha(R_t - \bar{R}_t)\pi_t(a), \qquad a \neq A_t.$$

Here $\bar{R}_t$ is a reward baseline, often the average reward so far. If the selected action gives reward above the baseline, its preference increases. If it gives reward below the baseline, its preference decreases.

### 9.1. Worked Example: Reward Above Baseline

Suppose there are three actions, $H_t(a_1) = H_t(a_2) = H_t(a_3) = 0$, so $\pi_t(a_i) = 1/3$. Let $\alpha = 0.1$, selected action $A_t = a_2$, reward $R_t = 2.0$, and baseline $\bar{R}_t = 1.0$. Then $R_t - \bar{R}_t = 1.0$.

$$\Delta H(a_2) = 0.1(1.0)(1 - 1/3) = 0.0667,$$

$$\Delta H(a_1) = \Delta H(a_3) = -0.1(1.0)(1/3) = -0.0333.$$

The selected action becomes more likely because it performed better than the baseline.

### 9.2. Worked Example: Reward Below Baseline

Suppose $\pi_t(a_1) = 0.5$, $\pi_t(a_2) = 0.3$, $\pi_t(a_3) = 0.2$, selected action $A_t = a_1$, reward $R_t = 0.4$, baseline $\bar{R}_t = 1.0$, and $\alpha = 0.1$. Then $R_t - \bar{R}_t = -0.6$.

$$\Delta H(a_1) = 0.1(-0.6)(1 - 0.5) = -0.03,$$

$$\Delta H(a_2) = -0.1(-0.6)(0.3) = 0.018, \qquad \Delta H(a_3) = -0.1(-0.6)(0.2) = 0.012.$$

The selected action becomes less likely because it performed worse than the baseline.

## 10. Contextual Bandits and Associative Search

The bandit problem discussed so far is **nonassociative**: the learner repeatedly faces the same decision situation. A contextual bandit introduces a context, or situation description, before the action is selected. The learner must associate different contexts with different useful actions.

> **Contextual bandit.** At each time step, the learner observes a context $x_t$, selects an action $A_t$, and receives reward for that selected action. The objective is to learn which actions work well in which contexts. This is closely related to **associative search**.

A contextual bandit is not yet the full reinforcement learning problem. The action affects the immediate reward, but it is not treated as changing the next context in a controlled sequential process. In full reinforcement learning, actions can affect future states and future rewards; therefore delayed consequences and long-term credit assignment become central.

### 10.1. Examples of Contextual Bandit Formulations

| Application | Context | Actions | Reward and reason it is contextual |
|---|---|---|---|
| Online advertising | User profile, device, query, page type, time of day. | Candidate advertisements. | Click, purchase, or revenue. The best advertisement depends on the observed user and page context. |
| News or content recommendation | User interests, recent browsing, location, and current session features. | Articles, videos, or posts. | Click, dwell time, or completion. Different users should receive different recommendations. |
| Clinical decision support | Patient record, symptoms, history, age, and risk indicators. | Candidate treatments or interventions. | Recovery indicator or improvement score. Treatment choice depends on patient-specific context. |
| Information retrieval | Search query, intent features, and previous interaction signals. | Ranking policy or selected result group. | Click satisfaction or relevance signal. The useful action depends on the query and user intent. |
| Adaptive tutoring | Learner profile, recent answers, and estimated mastery level. | Next exercise, hint, or explanation type. | Correctness or learning gain. The best instructional action depends on the learner's current context. |

These examples are contextual bandits because the learner receives side information before acting, but receives reward only for the chosen action. They are intermediate between ordinary bandits and full reinforcement learning: more expressive than a single-situation bandit, but still simpler than a multi-state sequential decision process.

## 11. Review Questions

1. Define stationary and nonstationary reward distributions in a bandit problem. Give one example of each and explain why it fits the definition.
2. Starting from $Q_{n+1} = \frac{1}{n}\sum_{i=1}^{n} R_i$, derive the incremental form $Q_{n+1} = Q_n + \frac{1}{n}[R_n - Q_n]$. Explain the role of the target and the estimation error.
3. An action has current estimate $Q = 5.0$. The next reward is $R = 9.0$. Compute the updated estimate using (a) sample-average step-size $1/n$ with $n = 4$, and (b) constant step-size $\alpha = 0.25$.
4. For the same action, suppose $Q_3 = 4.0$ and the next three rewards are $8, 8, 8$. Compute three updates using sample-average step-sizes $1/3$, $1/4$, and $1/5$. What happens to the size of the correction?
5. Explain why a constant step-size is more suitable than a $1/n$ step-size when reward distributions change over time.

6. Explain optimistic initial values. Why can they cause exploration even under greedy action selection? Why is this not a general exploration method?
7. For $t = 40$, $c = 2$, and three actions with $(Q, N)$ values $(1.2, 20)$, $(1.0, 5)$, and $(0.8, 1)$, compute the UCB score for each action and identify the selected action.
8. A streaming platform tests four recommendation widgets. User preference changes during a sports event. Should the learner use ordinary sample averages or a method with continuing responsiveness? Justify using the idea of stationarity.
9. A hospital decision-support system observes patient age, symptoms, and risk level before recommending one of several treatment options, and receives an outcome score later. Is this an ordinary bandit, contextual bandit, or full reinforcement learning problem under a one-step formulation? Justify.
10. A website can choose one of five homepage banners without observing any user-specific features. It receives reward 1 for sign-up and 0 otherwise. Model this as a bandit problem by identifying arms, rewards, and the missing feedback.
11. **Optional.** In a gradient bandit algorithm with two actions, suppose $\pi_t(a_1) = 0.7$, $\pi_t(a_2) = 0.3$, $A_t = a_1$, $R_t = 0.5$, $\bar{R}_t = 1.0$, and $\alpha = 0.1$. Compute the preference updates for both actions.

## 12. Points to Remember

- Stationary bandits have fixed reward distributions; nonstationary bandits have reward distributions that change over time.
- Incremental updates compute sample averages without storing all past rewards.
- The general update form moves an old estimate toward a target by a step-size-controlled amount.
- The step-size $1/n$ is natural for stationary sample averages; a constant $\alpha$ is useful for tracking nonstationary rewards.
- Optimistic initial values encourage early exploration but do not provide sustained exploration.
- UCB explores by adding an uncertainty bonus to the current action-value estimate.
- Gradient bandit algorithms learn action preferences rather than action-value estimates.
- Contextual bandits learn which actions work well in which contexts, but they do not model long-term state transitions as full RL does.

## 13. Required Reading

Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press. Required reading: Chapter 2, Sections 2.4 to 2.9.

## 14. References


Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). Cambridge, MA: MIT Press.

Bouneffouf, D., Rish, I., and Aggarwal, C. (2020). *Survey on Applications of Multi-Armed and Contextual Bandits.* 2020 IEEE Congress on Evolutionary Computation (CEC), 1–8. DOI: 10.1109/CEC48606.2020.918


## 15. Acknowledgements


These lecture notes are based on and adapted from Chapter 2 of Sutton and Barto's *Reinforcement Learning: An Introduction*, Second Edition.


*End of Topic -2 (Part 2)*